Hidde van de Beek

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PROFILE

PhD candidate in Pharmacometrics and Statistics with a background in methodology and statistics. I develop statistical methods and open-source software for population PK/PD modelling, with a focus on aggregate data modelling and exact derivatives for nonlinear mixed-effects models. Developer of `nlmixr2` and `ferx`, author of the R package `admixr2`.

WORK

PhD Pharmacometrics and Statistics 2024 – 2028
Leiden University, Leiden Academic Centre for Drug Research (LACDR)

- Developing statistical methods for the analysis of pharmacokinetic and pharmacodynamic data: aggregate data modelling, covariate model development, and exact standard errors for FOCE/FOCEI.
- Developer of the open-source NLME tools `nlmixr2` and `ferx`.

Teaching Assistant 2021 – 2024
Utrecht University and University Medical Centre Utrecht

Taught statistics courses from beginner to advanced level for Bachelor's programmes in the Faculty of Social Sciences, and later for the Bachelor's in Biomedical Sciences and the Master's in Epidemiology.

PUBLICATIONS AND SOFTWARE

From exact gradients to exact standard errors: third-order sensitivity equations for the FOCE and FOCEI population likelihood 2026
van de Beek H, Beldjenna M, Fidler ML, Zwep LB, van Hasselt JGC
bioRxiv preprint

doi:10.64898/2026.07.09.737434. Closed-form observed information for FOCE/FOCEI, implemented in `nlmixr2`.

admixr2: aggregate data modelling in R (formerly admr) 2025
Journal of Pharmacokinetics and Pharmacodynamics

doi:10.1007/s10928-025-10011-w. R package with a novel Iterative Reweighting Monte Carlo algorithm (3 to 100-fold speed-up). GitHub, CRAN.

nlmixr2 and ferx ongoing
Open-source software

Developer of nlmixr2 (R) and ferx (NLME engine with FOCEI, SAEM, analytical PK and ODE models).

EDUCATION

MSc Methodology and Statistics for the Behavioural, Biomedical and Social Sciences 2022 – 2024
Utrecht University

Key modules: Biomedical Statistics, Computational Inference with R, Bayesian Statistics, Multilevel Modelling, Causal Inference, Structural Equation Modelling. Thesis: *Decoding Predictive Performance: A Simulation Study on Information Criteria vs. Internal Performance Measures* (supervisors Dr. Maarten van Smeden and Dr. Anne de Hond), 8/10.

BSc Psychology 2018 – 2022
Utrecht University

Minor in Methodology & Statistics and Artificial Intelligence. Key modules: Mathematics for AI, Missing Data Theory and Causal Effects, Structural Equation Modelling, Adaptive Systems, Cognition. Thesis on consumer-grade fitness tracker owners and physical activity monitoring, with Dr. Vera Toepoel and Statistics Netherlands (CBS), 8/10.

Gymnasium, Natural Sciences profile 2012 – 2017
Utrechts Stedelijk Gymnasium

Physics, Chemistry, Biology, Math. Honourable participation in the National Physics Olympiad.

ADDITIONAL EXPERIENCE

Rower, student athlete 2018 – 2024
U.S.R. Triton

Trained 10 to 12 times a week at high-performance centres in Utrecht and Amsterdam. Regional Talent status, several National Championships and the Henley Royal Regatta (Temple 8+).

Coach 2019 – present
U.S.R. Triton

Coaching rowers of all levels on skill development, performance and team dynamics.

SKILLS

Languages: English, Dutch. **Software:** R, `nlmixr2`, `rxode2`, Quarto, Git.